

EasyEyes: Crowded dynamic fixation for online psychophysics

Fengping Hu

Department of Psychology, New York University,
New York, NY, USA



Joyce Y. Chen

Department of Psychology,
University of Michigan–Ann Arbor, Ann Arbor, MI, USA



Denis G. Pelli

Department of Psychology, New York University,
New York, NY, USA
Center for Neural Science, New York University,
New York, NY, USA



Jonathan Winawer

Department of Psychology, New York University,
New York, NY, USA
Center for Neural Science, New York University,
New York, NY, USA



Online vision testing enables efficient data collection from diverse participants, but often requires accurate fixation. When needed, fixation accuracy is traditionally ensured by using a camera to track gaze. That works well in the laboratory, but tracking during online testing with a built-in webcam is not yet sufficiently precise. Kurzawski, Pombo, et al. (2023) introduced a fixation task that improves fixation through hand-eye coordination, requiring participants to track a moving crosshair with a mouse-controlled cursor. This dynamic fixation task greatly reduces peeking at peripheral targets relative to a stationary fixation task, but does not eliminate it. Here, we introduce a crowded dynamic fixation task that further enhances fixation by adding clutter around the fixation mark. We assessed fixation accuracy during peripheral threshold measurement. Relative to the root mean square gaze error during the stationary fixation task, the dynamic fixation error was 55%, whereas the crowded dynamic fixation error was only 40%. With a 1.5° tolerance, peeking occurred on 7% of trials with stationary fixation, 1.5% with dynamic fixation, and 0% with crowded dynamic fixation. This improvement eliminated implausibly low peripheral thresholds, likely by preventing peeking. We conclude that crowded dynamic fixation provides accurate gaze control for online testing.

Introduction

Visual psychophysics often requires that the experimenter know where the participant is looking.

Often, the participant is instructed to keep their eyes on a fixation mark. Ideally, the experimenter verifies fixation accuracy with a gaze tracker. This is especially important for eccentricity-dependent tasks, like acuity (Weymouth, 1958) and crowding (Bouma, 1970). Accurate gaze tracking, however, requires expensive equipment and in-laboratory testing. This excludes laboratories without the proper equipment and restricts the participant pool to the local population. This population is often WEIRD (Western, Educated, Industrialized, Rich, Democratic) and not representative of the global population (Henrich, Heine, & Norenzayan, 2010; Rad, Martingano, & Ginges, 2018).

Online testing solves multiple problems. With no need for a laboratory, one can test a larger, more representative sample with far less effort. However, vision scientists have often missed out on this opportunity because of concerns about monitoring fixation. Several open-source packages are available for doing gaze tracking online using the computer's webcam (Papoutsaki et al., 2016; Xu et al., 2015). However, software using the typical resolution of a built-in webcam limits the gaze-tracking accuracy to around 4° (Papoutsaki et al., 2016; Semmelmann & Weigelt, 2018). Recently, Kaduk, Goeke, Finger, and König (2024) reported that adding a high-resolution webcam can decrease the gaze tracking error to 1.4° (Kaduk et al. 2024), but this technique requires buying the equipment for the participants, and frequent calibration. Alternatively, Kurzawski, Pombo et al.

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(2023) showed that asking participants to use a cursor to track a moving fixation crosshair (dynamic fixation) yielded good fixation in most participants, meaning they kept their eyes fixed on the moving crosshair. We use the word “fixation” here to mean keeping the eyes on the crosshair, whether or not it is moving. The crosshair moves slowly and predictably, on a circular path, to make it easy to pursue. We wondered whether we can improve the method to achieve good fixation in all participants, eliminating the need for an eye tracker.

To understand our motivation, it helps to consider the fixation task together with the principal task of the experiment. Kurzawski, Pombo et al. (2023) used the dynamic fixation task to measure a peripheral threshold. A trial begins with central fixation, followed by a brief flash of the stimulus, a three-letter triplet, in the periphery. Without an effort to suppress saccades, participants tend to move their eyes to the stimulus. We call this peeking. The stimulus is flashed briefly, to disappear before the participant’s eyes can reach it. Nonetheless, the participant knows a stimulus will appear, and might make an anticipatory eye movement, much as a soccer goalie sometimes dives to the left or right in anticipation of the penalty kick. Even if the eye movement only approaches the target by chance, it will sometimes land near the target, improving performance and corrupting the threshold estimate (Kurzawski, Burchell et al., 2023).

The participant feels the tension of opposing attractions to gaze. On the one hand, the dynamic fixation task encourages fixating the crosshair because, when tracking an object with your hand, it is natural to track with your eyes as well (Koken & Erkelens, 1992; Xia & Barnes, 1999). On the other hand, it is also natural to shift gaze toward an anticipated target. The current study sought to enhance the tendency to fixate, so it would always win.

To assess progress toward this goal, we compared three fixation tasks: stationary fixation, dynamic fixation, and crowded dynamic fixation. In stationary fixation, a static crosshair is displayed at the center of the screen, serving as a visual anchor for the participant’s gaze. This is the traditional way to present the fixation mark in psychophysical testing with peripheral stimuli. In dynamic fixation (Kurzawski, Pombo et al., 2023), participants are asked to use the mouse and mouse-controlled cursor to manually track a crosshair moving along a circular trajectory. In crowded dynamic fixation, the tracking task is enhanced by the addition of three distractors moving randomly near the moving crosshair. We implemented all three kinds of fixation while doing a peripheral threshold task and used a gaze tracker to measure fixation accuracy.

Our results validate the effectiveness of crowded dynamic fixation in maintaining stable central fixation during peripheral visual testing. Among the three types of fixation task, crowded dynamic fixation is the

most effective in keeping participants’ gaze at fixation. Crowded dynamic fixation provides this benefit with negligible increase in the experiment duration. The new crowded dynamic fixation task enables online testing with accurate fixation.

Methods

Participants

Twenty-one participants, aged 17 to 41 years, participated in our experiment. One participant was excluded because the Eyelink 1000 could not reliably track their eyes, and two participants were excluded from the crowding threshold analysis because of a software failure on that day. All participants were native English speakers, and, if they needed optical correction, they wore contacts. None of the participants had prior experience with psychophysical testing. Participants aged 18 years and older signed consent and debriefing forms. Participants younger than 18 years old signed assent and debriefing forms, and their parents or guardians signed consent forms.

Thirty-three participants, aged 19 to 61 years, participated in our online experiment via Prolific Academic (<https://prolific.co/>). One participant was excluded for being unable to do the task (crowding threshold estimated $>100^\circ$). All participants self-reported to have normal vision and use English as their primary language. All participants viewed and agreed to consent and debriefing forms.

All in-laboratory and online participants were compensated \$15/hour for their participation. Both the in-laboratory and the online experiments were approved by the New York University Institutional Review Board (IRB-FY2016-404).

Apparatus

The EyeLink 1000 eye tracker (SR Research, Ottawa, Ontario, Canada) with a tower-mount setup was used to track participants’ right eye throughout the experiment at a sampling frequency of 1 kHz. An Apple iMac 27” computer presented stimuli on an external monitor, LG 27” UltraFine 5K (27MD5KL-B), with a 59.5×33.5 cm screen with $5,120 \times 2,880$ pixels, and a frame rate of 60 Hz, and a luminance of about 275 cd/m^2 . The participant placed their chin on an adjustable chin rest 40 cm away from the monitor and used a mouse to perform the tasks.

EasyEyes

We used EasyEyes, an open-access web app for creating and running psychophysical experiments

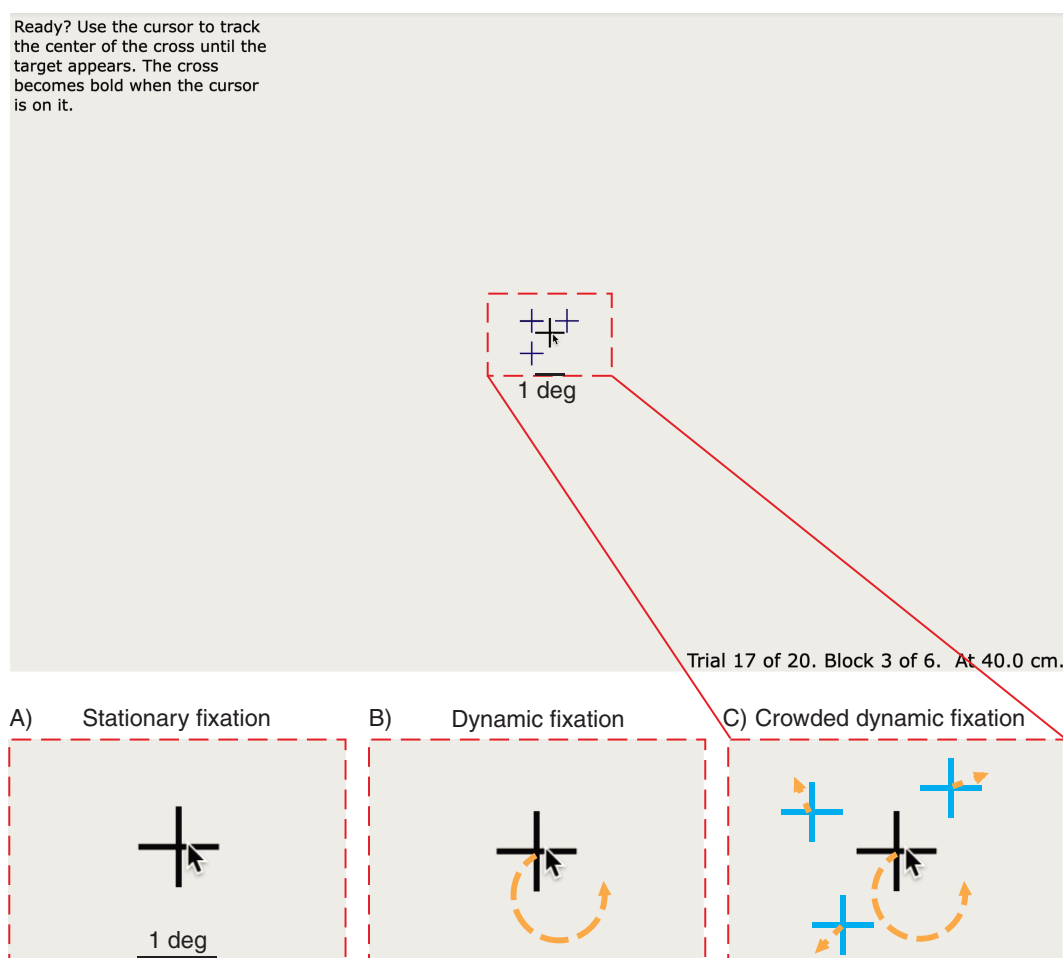


Figure 1. Tracking task. Participants are asked to keep the cursor on the crosshair. The red dashed rectangle indicates the size of the area depicted in A–C. (A) Stationary fixation: The crosshair is static. (B) Dynamic fixation: The crosshair moves along a circular trajectory at a constant speed (indicated by the orange arrow). (C) Crowded dynamic fixation: The moving crosshair (black) is surrounded by three distractors. For display purposes, the color difference between fixation and distractors is exaggerated.

(Jiang, Burchell, & Pelli, 2021). EasyEyes experiments are created by uploading a Microsoft Excel file, with one column per condition, specifying the desired parameters (such as stimulus size and duration). EasyEyes calibrates the screen size by asking the participant to indicate the size of a familiar object (such as a credit card) on the screen (Li, Joo, Yeatman, & Reinecke, 2020). EasyEyes uses the PsychoJS implementation of Quest (PsychoJS, n.d.; Watson & Pelli, 1983) to measure thresholds. It records the location of the participant's cursor and the fixation crosshair throughout the experiment at a frequency of 60 Hz. Gaze tracking data is recorded separately and integrated at the end of the experiment (see Data analysis).

Fixation task

Participants were asked to keep the cursor on the crosshair, which was either static or moving (described

below). The cursor is “on” the crosshair whenever the cursor tip is inside a 0.15° “hotspot radius” of the crosshair center. Cursor tracking feedback is provided by increasing the crosshair stroke thickness 1.5-fold to indicate that the cursor is on the crosshair. The 0.15° hotspot radius was chosen based on pilot experiments that included cursor tracking and gaze tracking to minimize cursor and gaze errors, and to limit the perceived difficulty (Supplementary Figure S1).

Stationary fixation

A static crosshair is displayed at the center of the screen (Figure 1A). The cross consists of a horizontal and a vertical black line, each 1° in length and 0.05° thick.

Dynamic fixation

The crosshair (same as above) moves counterclockwise at a speed of 0.4 deg/s along a circular trajectory. The trajectory centers at the center of the screen and

has a radius of 0.5° (Figure 1B). Each trial begins with the crosshair at a random point on this trajectory. The motion speed and motion radius of the crosshair were chosen to minimize tracking error, gaze error, and perceived difficulty (Supplementary Figure S1).

Crowded dynamic fixation

The crosshair motion is the same as for dynamic fixation, but three distractors are added near the crosshair (Figure 1C). The distractors were added to make it difficult to see the crosshair when looking away (i.e., breaking fixation), but not when fixating on it. Each distractor consists of a horizontal and a vertical dark blue line with a length of 0.8° and a thickness of 0.05° . Thus, the distractor crosshairs bear a family resemblance to the fixation crosshair, but have an obviously different color and size. The three distractors move in Brownian motion with an instantaneous speed of 1.1 deg/s within 1° of the crosshair. To avoid clustering, a repulsive displacement is added to each distractor. The repulsion is implemented using the standard gravitational formula, but with repulsion instead of attraction. We computed a repulsion direction and length from each pair and summed these to determine the displacement of each distractor. The direction of displacement for each pair is in the direction of the line connecting the pair, away from the other distractor. The length h of the displacement (in degrees) between distractors i and j is:

$$h_{i,j} = -g t d_{i,j}^{-2}, \quad (1)$$

where t is the frame duration (seconds), $d_{i,j}$ is the distance (degrees) between distractors i and j , and g is $0.25 \text{ deg}^3/\text{s}$.

Crowding task

Each trial consists of three intervals: Track, View, and Respond (Figure 2). During the track interval, participants are instructed to keep the cursor tip on the crosshair center. The tracking task continues until the participant has tracked continuously for the required duration (0.75–1.25 seconds, randomly selected for each trial). If tracking breaks (cursor leaves hotspot radius), then the trial resets and waits for tracking to resume, when a new random length interval is sampled. This tracking and resetting continues until the participant manages to track for a full (random length) interval. If a participant fails to initiate a trial after 30 seconds of tracking, the trial is skipped.

Upon completion of tracking, the trial proceeds to the view interval. The crosshair remains frozen on the screen at its final position, while the cursor

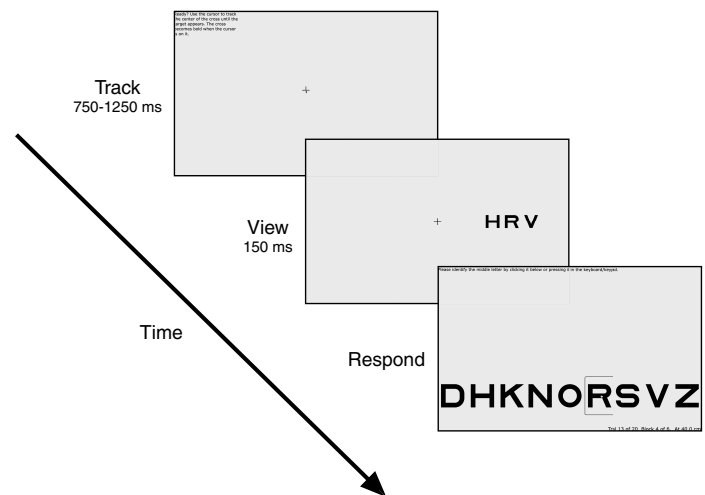


Figure 2. Crowding experiment procedure. (1) Track: During each trial, the participant is first asked to use a mouse to make the mouse-controlled cursor track the fixation crosshair. (2) View: After continuous successful tracking for 750 to 1,250 ms (random from trial to trial), a letter trigram is shown for 150 ms, centered $\pm 10^\circ$ from fixation (i.e., current crosshair location), on the horizontal meridian. The crosshair remains on the screen at its final position. (3) Respond: After 150 ms, the trigram disappears, and the participant identifies the target letter by pressing it on the keyboard or clicking one of the nine letters along the bottom of the screen.

and any distractors disappear. Simultaneously, a letter trigram appears for 150 ms, centered at $\pm 10^\circ$ eccentricity (relative to the current position of the fixation crosshair). The position of the trigram (left or right meridian) is random across trials. The size of the letters is 0.71 times the center-to-center spacing. Size and spacing covary across trials, controlled by the staircase. The three letters in the trigram are chosen randomly, without replacement, from nine letters, DHKNORSVZ, of the Sloan font (Pelli, Robson, & Wilkins, 1988; Sloan, 1959). The letter C was excluded because it is too easily confused with O (Elliott, Whitaker, & Bonette, 1990). The middle letter is the target.

The respond interval begins after the 150-ms view. The crosshair disappears and an array of all nine possible target letters appears at the bottom of the screen. The participant is asked to identify the target by pressing that key in the keyboard or clicking that letter on the screen.

Experimental procedure

Each participant completed two in-laboratory sessions on separate days. The first session consisted of a practice experiment and the first run of the full

experiment. The second session was the second run of the full experiment.

In the practice experiment, participants completed 6 blocks of 20 trials each. Each trial followed the general procedure of the crowding task as described, including the random left vs. right position of the target on each trial. However, for the first three practice blocks, only a single letter, rather than the letter trigram, was presented during the view interval. Participants were instructed to select the letter they saw during the respond interval. Fixation was stationary in the first block, dynamic in the second, and crowded dynamic in the third. These initial blocks were designed to familiarize participants with the general experimental procedure and the three fixation tasks. The final three practice blocks were the same except that they included the letter trigrams.

The main experiment began with calibrating the eye tracker. The experimental session consisted of 3 blocks of 70 trials each. Within each block, the fixation condition was constant, either stationary, dynamic, or crowded dynamic, with the order randomized. Each 70-trial block consisted of two randomly interleaved 35-trial staircases, one for right and one for left. The center-to-center letter spacing was controlled by QUEST (Watson & Pelli, 1983). The second session was identical to the first, except for no practice and with new random ordering of blocks. At the end of each block, QUEST estimates a threshold for each condition (left or right) as the 53rd quantile of the posterior probability density function of threshold vs. spacing.

Online experiment

The online experiment measured crowding on the right meridian at 10° eccentricity. Participants first completed six practice blocks, which were identical to the practice session for the in-laboratory experiment, but without left stimulus trials (i.e., each practice block was shortened to 10 trials, with the stimulus appearing on the right meridian for every trial). Then, they completed a set of main experiment blocks, which were identical to the main experiment for the in-laboratory experiment, except that each block contained only one 35-trial staircase for measuring the right crowding threshold. In total, they completed six blocks, including test and retest of the three fixation conditions.

Data analysis

EasyEyes–Eyelink time alignment

EasyEyes records crosshair and cursor position (in each frame, at 60 Hz), while a MATLAB program records the gaze position (at 100 Hz) measured by

EyeLink. Both programs save the data as CSV files including POSIX timestamps. The POSIX timestamps are used to align the two data files.

Gaze correction

Despite calibration of the eye tracker before the experiment, there is some drift in calibration over the course of the experiment. For each trial, we correct for this error by calculating the average gaze position in the last 400 ms of the track interval and subtracting that from the measured gaze position in every frame of that trial.

To remove blinks from the eye tracking data, we first identify blinks by calculating the speed and velocity of gaze. A blink is detected when the velocity exceeds 1,000 deg/s or the gaze position deviates more than 20° from fixation (likely off screen). The start and end frames of each blink are then marked, and the gaze positions of the entire blink period, along with an additional 100 ms before and after, are replaced with NaN.

Threshold estimation after combining test and retest

Increasing the number of trials given to the QUEST staircase increases the precision of the threshold estimates. For both the in-laboratory and online experiments, we calculated crowding thresholds after combining two 35-trial staircases per condition.

Results

Crowding thresholds were measured at $\pm 10^\circ$ eccentricity on the horizontal meridian using each of three fixation tasks: stationary, dynamic, and crowded dynamic. The crowding triplet appeared randomly on either the left or the right meridian. This was implemented by randomly interleaving two staircases, one for left and another for right. Gaze position was monitored by an eye tracker.

Crowded dynamic fixation reduces gaze deviation throughout the trial

We first demonstrate the advantage of the crowded dynamic fixation task by examining gaze deviation at each time point in a trial. We do so by plotting gaze position during every trial, grouped by fixation condition, for an example participant (Figure 3). The traces indicate the horizontal (left column) or vertical (right column) distance between crosshair and gaze. The meridian of the crowding triplet is indicated by color (red for right, green for left).

With stationary fixation (Figure 3A), this participant made saccades (i.e., eye movements $\geq 1^\circ$) during 10 of 70 trials in either the track or view interval. These saccades are problematic for measuring peripheral thresholds because they change the target eccentricity from the nominal 10° . These saccades were always on the horizontal axis, typically only a few degrees, much less than the 10° stimulus eccentricity, and in a direction (left vs. right) uncorrelated with the unpredictable left-or-right target location. The timing, amplitude, and direction of the saccades indicate that the participant had some expectation of when, though not on which side, the stimulus would appear. We refer to saccades initiated during the track or view interval as anticipatory saccades. To confirm that the large bias toward horizontal saccades over vertical was indeed owing to participant expectation about the target locations rather than a general bias for horizontal, we collected one additional dataset, identical to the main experiment except that the targets were $\pm 10^\circ$ on the vertical axis rather than horizontal axis. In this experiment, the bias in anticipatory saccades switched to the vertical axis, as expected (Supplementary Figure S2). In contrast, during the respond interval,

the participant usually made a saccade to the target location (10° amplitude, correct side). These saccades were clearly in response to seeing the target, and inconsequential because the stimulus disappeared before the saccade began. We refer to these as response saccades. A key goal of this project was to minimize anticipatory saccades. Response saccades do not matter. Because anticipatory saccades are away from the target one-half of the time, they do not dramatically reduce crowding thresholds. However, our simulations show a small expected reduction in threshold from peeking (Kurzwski, Burchell et al., 2023).

The frequency and amplitude of anticipatory saccades decreased with dynamic fixation, as reported by Kurzwski, Pombo et al. (2023). However, there were still four anticipatory saccades. The crowded dynamic fixation eliminated anticipatory saccades. There were no vertical saccades in any of the three fixation conditions.

Improvement in fixation accuracy from stationary to dynamic to crowded dynamic is shown by summarizing fixation accuracy at each time point in the trial. Specifically, we calculated the root mean square distance between the crosshair (i.e., fixation mark) and gaze (root mean square error [RMSE]), across the 70 trials

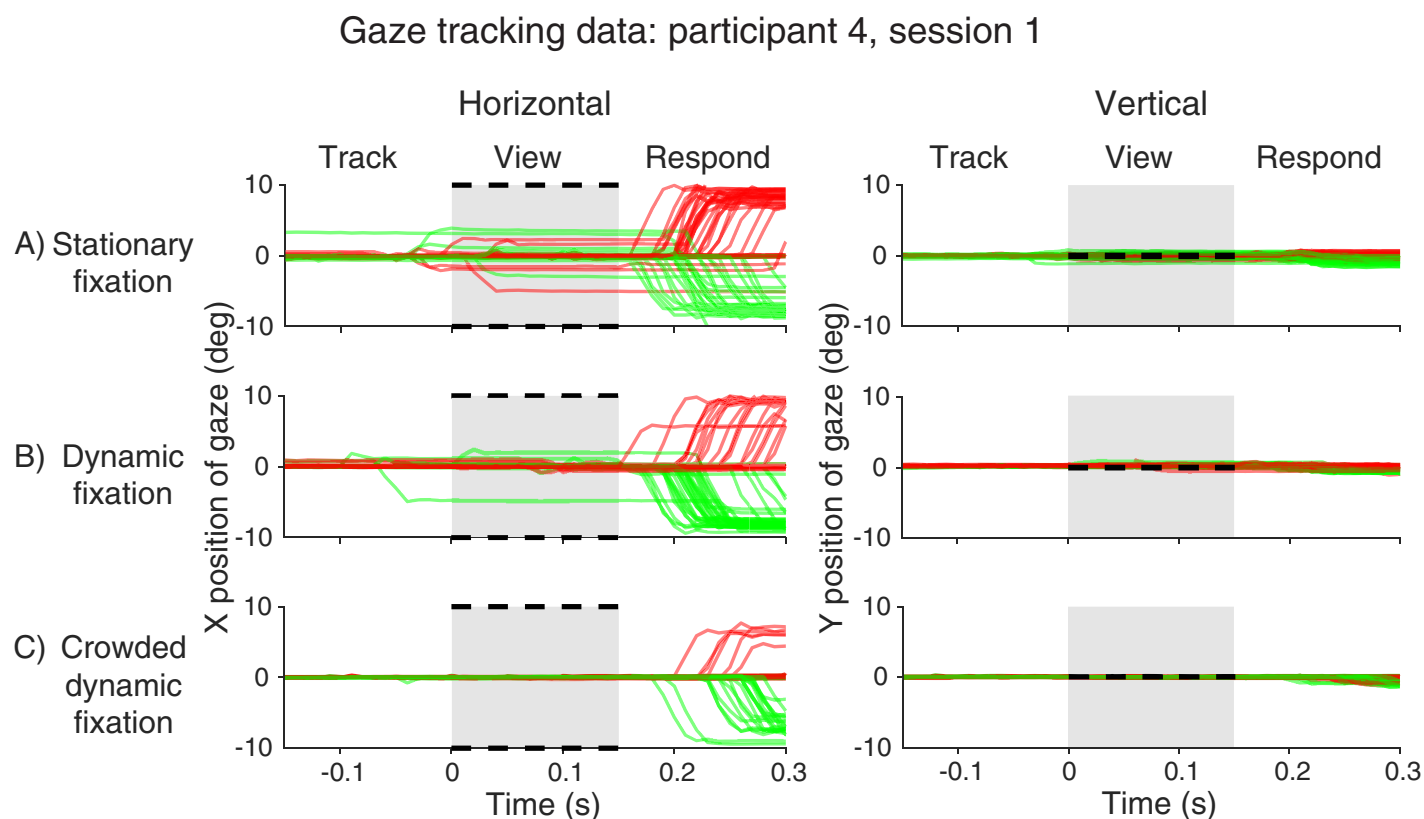


Figure 3. Gaze data for participant 4. (A) Horizontal and vertical gaze position, relative to the crosshair position, during stationary fixation trials. The gray patch indicates target presence on the screen. Black dotted lines mark the two possible target locations ($\pm 10^\circ$ horizontally). Each red or green trace is gaze position over time during one trial. Red traces are for trials with right targets and green traces are for trials with left targets. (B) Gaze position during dynamic fixation trials. Plotted gaze position is relative to current crosshair position. (C) Gaze position during crowded dynamic fixation trials (plot_all_trials_xy.m).

per fixation condition, separately for each time point (video frame) within the trial. This summary metric is shown for one participant (Figure 4A) and the group average (Figure 4B). The important part of the trial is the view interval, when the stimulus is on the screen (shaded gray region). During the view interval, the RMSE is highest for stationary fixation, intermediate for dynamic fixation, and lowest for crowded dynamic fixation, for both the example participant (Figure 4A) and the group average (Figure 4B). The data are replotted with RMSE calculated relative to stationary fixation in Figures 3C and 3D. For the group average, the RMSE for dynamic fixation is approximately 55% and crowded dynamic fixation is approximately 40%, relative to stationary fixation.

Crowded dynamic fixation reduces the number of trials with large eye movements

The previous section reported the benefit of crowded dynamic fixation using a continuous measure (RMSE gaze deviation over time). Here we quantify the benefit with a discrete measure, the number of trials with large gaze deviations, a common metric of gaze behavior for psychophysical experiments.

Crowded dynamic fixation greatly reduced the number of trials with large gaze deviations, or fixation breaks. We calculated the number of such trials using several different tolerances for a fixation break (Figure 5). The trial is classified as having broken fixation if the participant's fixation error (gaze distance

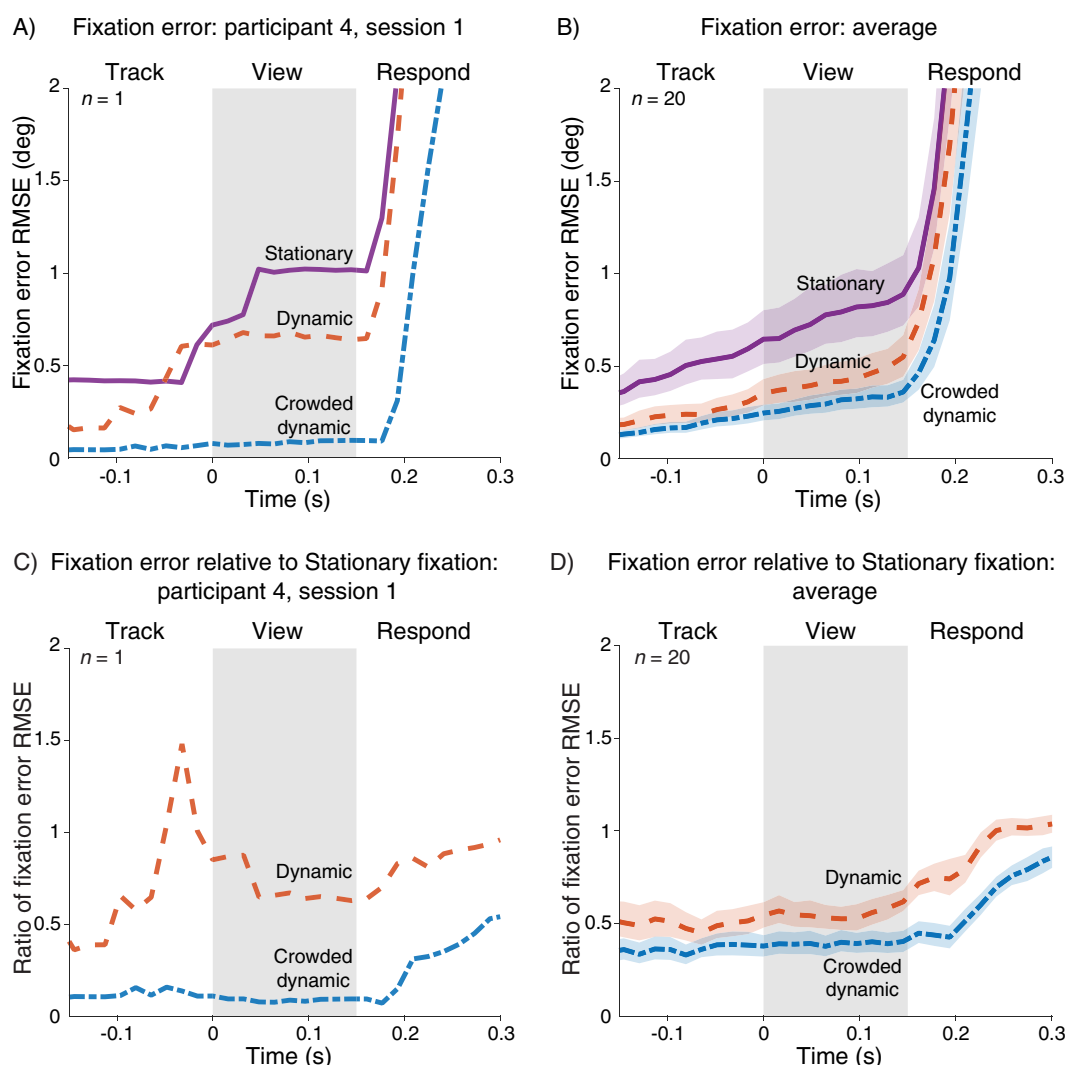


Figure 4. Fixation error for three fixation tasks. (A) Distance between gaze position and crosshair position for participant 4, session 1. Each curve represents the RMSE over the 70 trials for that condition at each time point (plot_rmse_single_participant.m). (B) Geometric means of fixation error RMSE from 40 experiment sessions (20 participants, 2 sessions each) (plot_rmse_over_time_group.m, bootstrap_plot_rmse_group.m). (C and D): Same as (A) and (B), but relative to the RMSE fixation error during stationary fixation (plot_rmse_single_participant_ratio.m, bootstrap_plot_rmse_group_ratio.m). For (B) and (D), shaded regions indicate 68% confidence intervals bootstrapped across participants.

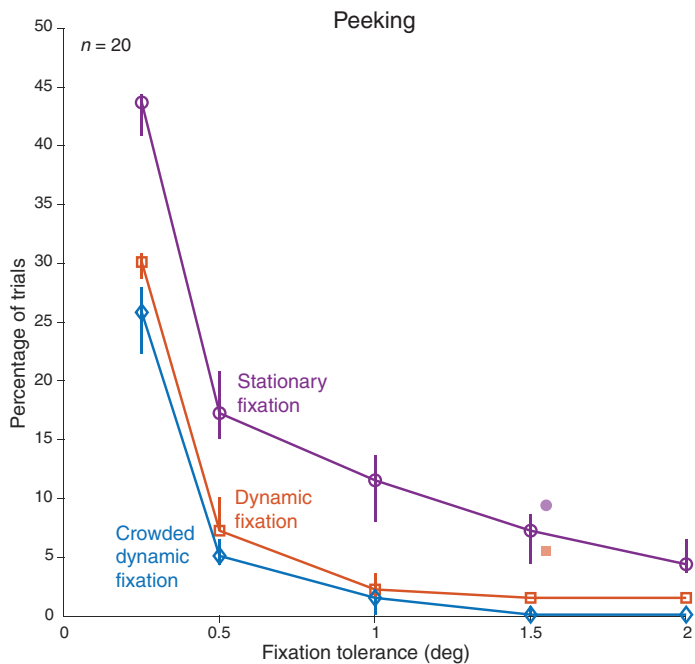


Figure 5. Frequency of fixation breaks. For each fixation condition, with several different fixation-error tolerances (x axis), fixation breaks were counted and shown as a percentage of total trials (y axis). Each open symbol on the curve represents the median percentage of trials, calculated from 40 sessions, where participants broke fixation during triplet presentations. Error bars indicate 68% confidence intervals bootstrapped across sessions. Filled symbols: Kurzawski, Pombo et al. (2023), horizontally offset by 0.05° for display purposes. (plot_peek_number_sweep.m and plot_peeking_summaries.m).

from crosshair) exceeds tolerance at any time during the view interval. With the 1.5° tolerance, fixation breaks are reduced from 7% (5 of 70 trials) with stationary fixation to 1.5% (1 of 70) with dynamic fixation, to none with crowded dynamic fixation. This reduction from stationary to dynamic is comparable with the results reported by Kurzawski, Pombo et al. (2023). The order is the same at every tolerance: the largest number of fixation breaks for stationary fixation, and the fewest for crowded dynamic fixation.

Crowded dynamic fixation eliminates abnormally small crowding thresholds that can be caused by peeking

The main goal of this study is improved fixation. Here we consider how the improved fixation affected the measured peripheral thresholds. Figure 6 summarizes crowding thresholds estimated with each fixation task. The key in this figure is the reduced range of the crowding thresholds in the crowded dynamic condition (Figure 6A). Kurzawski, Pombo et al. (2023)

measured Bouma factors of 50 participants with and without a gaze-contingent procedure, and found that the gaze-contingent procedure increased the minimal Bouma factor. Among the 200 Bouma factors the authors obtained at 10° eccentricity on the left and right horizontal meridians (50 participants, tested twice on each meridian), the minimum Bouma factor was 0.096. We take this Bouma factor as the lowest expected Bouma factor when there is no gaze error (i.e., no peeking). On this basis, we call thresholds lower than this abnormally low. We find that with stationary and dynamic fixation, 3.5% of the crowding thresholds are below the expected minimum (Figure 6A, top and middle).

We consider three possible causes of abnormally low thresholds.

1. The observer might just have a very low crowding threshold. This is unlikely for two reasons. First, in a larger dataset with a gaze-contingent experiment (Kurzawski, Burchell et al., 2023), we never found Bouma factors below 0.096, meaning that, if they occur, they would be very rare. Second, no observer has an abnormally low threshold in all three fixation conditions.
2. The threshold estimation procedure, QUEST, produces a distribution of threshold estimates given a true threshold, and sometimes the measurement will fall in the tail of that distribution, but that would be hard to repeat. This is unlikely. First, this cannot explain the difference between fixation conditions. Second, when we recompute the threshold from the concatenation of test and retest trials, which makes the estimate more precise, diminishing the tails. We still observe abnormally low thresholds in the Stationary and Dynamic conditions (Supplementary Figure S3).
3. The participant peeks. This is the most likely explanation, partly because the dynamic and the crowded dynamic fixation methods were designed to reduce peeking, and indeed, we observed lower gaze deviations.

We conclude that crowded dynamic fixation greatly reduces peeking (Figure 5) and abolishes these abnormally small thresholds (Figure 6A, bottom).

Shifting our attention from the tail to the center of the distributions, there is a small difference in the average crowding thresholds between conditions (Figure 6B). Examining only the 5 out of 36 experimental sessions without fixation breaks (peeking), the crowding thresholds do not differ across the three fixation conditions (Figure 6C). In all three conditions, we observe the expected left–right asymmetry, confirming prior findings of a right-field advantage (Kurzawski, Burchell et al., 2023).

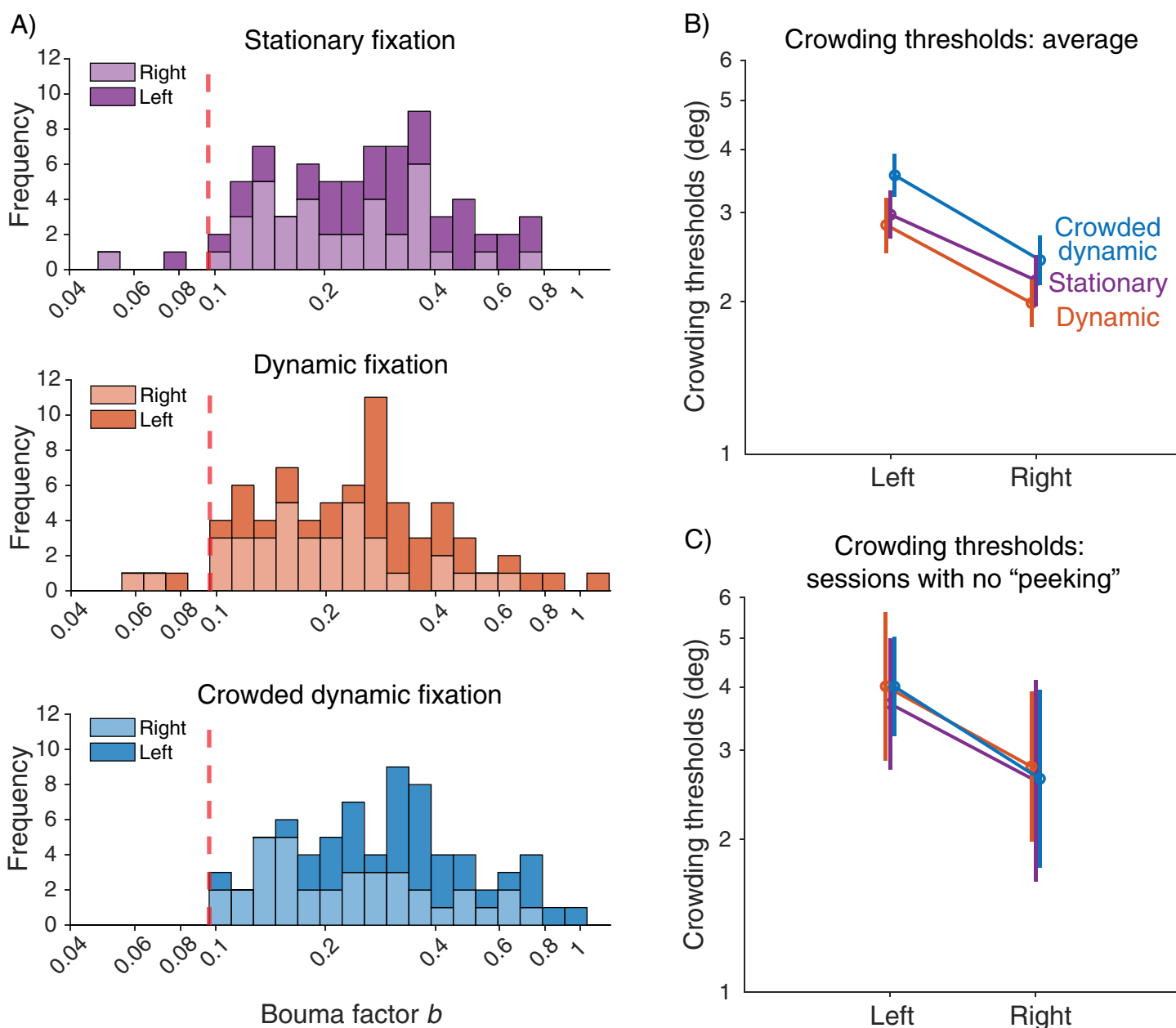


Figure 6. Crowding thresholds measured with the three fixation tasks. (A) Stacked histograms of all Bouma factors acquired under each fixation condition ($n = 72$ for each condition, 18 observers $\times$ 2 locations $\times$ 2 repeats). Bouma factors to the left of the red dashed line ($b = 0.096$) are considered abnormally small (plot_thresholds.m). (B). Geometric mean of crowding thresholds for 19 participants (plot_thresholds.m). (C) Geometric mean of crowding thresholds for the five sessions (three participants) with no fixation breaks (tolerance = 1.5°) in any condition. For (B) and (C), the data points are jittered horizontally for visibility (plot_thresholds_obedient_participants.m).

With crowded dynamic fixation, manual tracking is difficult while averting gaze

Distractors were included in crowded dynamic fixation so that gaze aversion would result in crowding that would challenge manual tracking. We verified that gaze aversion worsens manual tracking by asking 4 participants to manually track the crosshair while

fixating on a red dot that was 1° , 2° , 4° , or 8° displaced from the crosshair (Figure 7A). When participants fixated farther away from the crosshair, cursor tracking error increased, and it increased more if the distractors were present (crowded dynamic fixation) than if the distractors were absent (dynamic fixation) (Figure 7B). The inset to Figure 7B shows an example of the large decline in manual tracking accuracy when fixating a

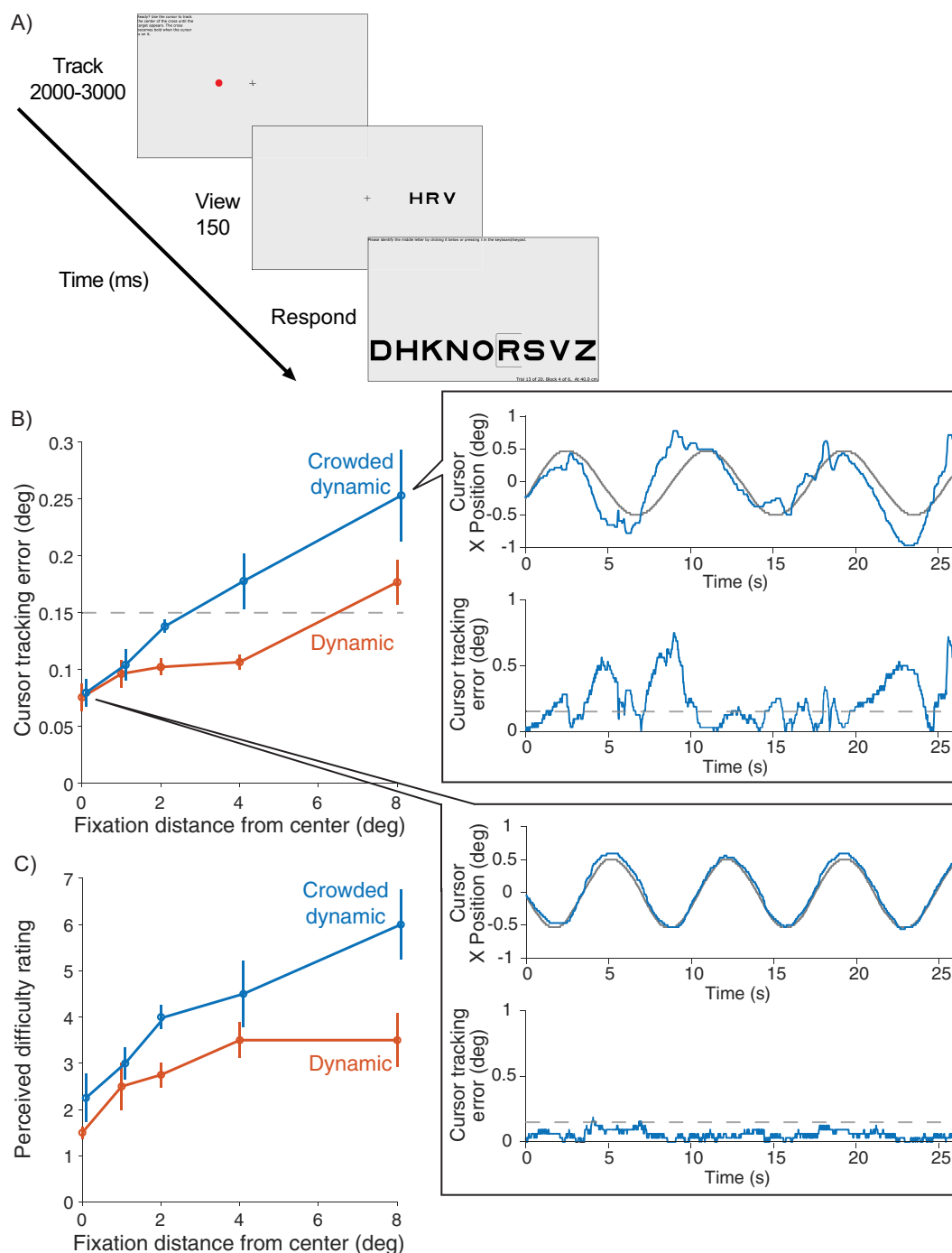


Figure 7. Cursor tracking accuracy while fixating elsewhere ($N = 4$). The purpose was to determine how well participants could track peripherally. **(A)** The fixation task was either dynamic or crowded dynamic (Figure 1). In both cases, participants tracked the moving crosshair with the cursor. While doing so, they either fixated the crosshair (0 distance) or a red dot that was 1°, 2°, 4°, or 8° displaced from the crosshair. **(B)** Cursor tracking error as a function of the distance between fixation and the crosshair (flies_noFlies_dot.m). The blue curve is shifted horizontally to avoid occlusion. Blowout plots cursor tracking data from an example participant in the crowded dynamic condition when the person fixated 8° from the screen center (top blowout) and fixate at the crosshair (bottom blowout). The top figure of each blowout plots X positions of cursor (blue curve) and crosshair positions (gray curve) over time. The bottom figure of each blowout plots cursor tracking error over time (blue curve) and the dashed gray line marks the 0.15° hotspot radius (plot_tracking_demo.m). **(C)** Perceived difficulty rating for the same task as **(B)** (flies_noFlies_rating.m).

dot 8° away from the crosshair. In addition, compared with dynamic fixation, participants reported feeling more challenged with crowded dynamic tracking when required to look away (Figure 7C). However, when participants were foveating the crosshair, cursor tracking error was equally low for both dynamic and crowded dynamic tracking, as was perceived difficulty. This finding shows that adding the distractors does not add tracking difficulty when participants maintain central fixation. This result is expected for crowding, which is generally negligible at fixation and grows linearly with eccentricity (Bouma, 1971; Rosen, Chakravarthi, & Pelli, 2014).

It might seem that participants can get away with 1° of gaze deviation because the average cursor error is less than the hotspot radius. However, our gaze data in the main experiment showed that participants did not do this (Figure 5). This might be because of the way we enforced tracking. Participants were required to continuously track within a hotspot radius for approximately 1 second. Any deviation beyond the hotspot resets the timer. Even if the average cursor error while fixating 1° away remains less than the hotspot radius, cursor errors beyond the hotspot radius become more common, prolonging the trial.

Crowded dynamic fixation slightly increases experiment duration

The two dynamic fixation tasks slightly lengthen the experiment because they add a tracking interval to each trial before stimulus onset. Figure 8 shows the impact of the fixation task on several durations. Tracking duration per trial increases (Figure 8A), but viewing (150 ms, not shown) and response duration (Figure 8B) are unchanged, resulting in slightly increased duration of the 70-trial block (Figure 8C): 5.5 minutes for

stationary, 6.2 minutes for dynamic, and 6.5 minutes for crowded dynamic. Thus, an experiment with crowded dynamic fixation costs 5% longer than one with dynamic fixation, and 18% longer than one with stationary fixation.

Crowding thresholds measured online

The above in-laboratory studies used gaze tracking to assess gaze accuracy. However, our ultimate goal was to offer a method of online testing, which has access to broader populations. So we measured crowding thresholds using the three fixation methods with online participants. Because we do not measure gaze position during online testing, the only measure of success is indirect; namely, that the crowding thresholds are not abnormally low. In the laboratory, we found 3.5% of thresholds to be low enough to suggest peeking with stationary and dynamic tracking. One reason for the low percentage is that peeking does not translate directly to lower thresholds, as peeking is anticipatory and will lead to saccades away from the target on about one-half of the trials.

Given this low percentage, repeating the same experiment online would not be a sensitive test of the benefit of crowded dynamic fixation. To make the link between peeking and thresholds more direct, we eliminated target location uncertainty by always placing the target at the same location (10° to the right of fixation). Target certainty makes participants more likely to peek, and increases the peeking benefit. Indeed, Figure 9 shows that 28% of crowding thresholds with stationary fixation were abnormally low (9 of out 32), compared with 2 of 32 with dynamic and 0 of 32 with crowded dynamic fixation. The effect of peeking is also evident in the differences in standard deviation between the three fixation conditions (0.48,

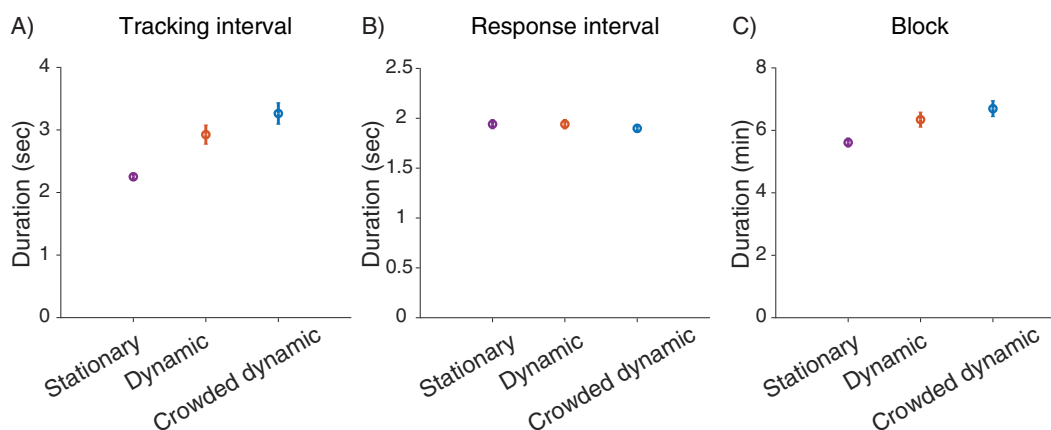


Figure 8. Effect of fixation task on experiment duration (20 participants, 2 sessions each). (A) Average tracking interval duration. (B) Average response interval duration. (C) Average duration of 70-trial block. Error bars represent $\pm$ standard error of the mean (plot_time_spent.m).

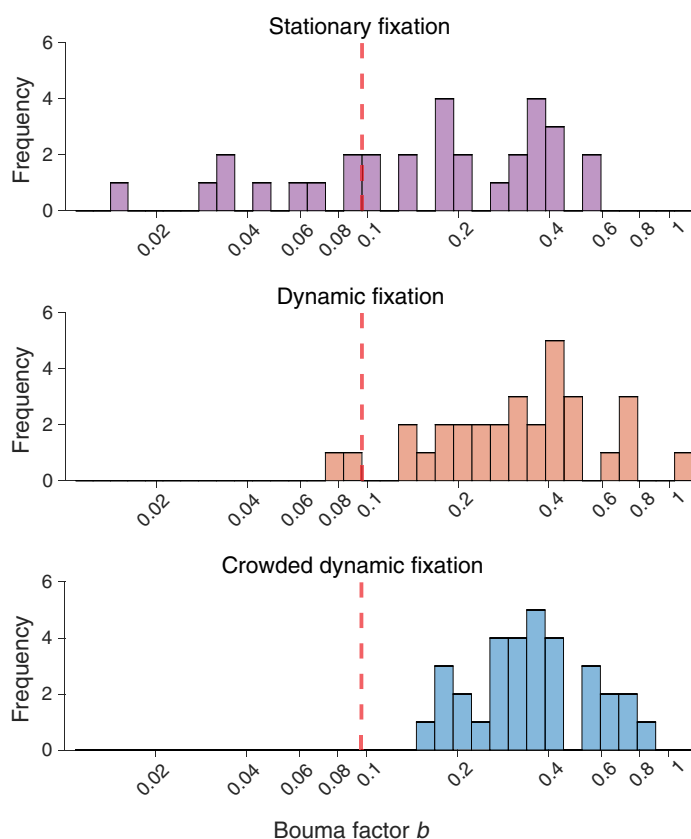


Figure 9. Crowding thresholds measured with the three fixation tasks in an online experiment. Histograms of all Bouma factors acquired under each fixation condition ($n = 32$ for each condition). In this experiment, the stimulus always appeared in the right meridian. Bouma factors to the left of the red dashed line ($b = 0.096$) are considered abnormally small (combine_quest_staircases_online.m).

0.33, and 0.20 for stationary, dynamic, and crowded dynamic fixation, respectively), which is independent of the red line. The tighter threshold distributions indicate improved fixation in the two dynamic tracking tasks compared with the usual stationary fixation.

There is a numerical advantage of crowded dynamic over dynamic fixation, but the numbers are too low for statistical testing (1 vs 0 thresholds below our criterion). Note that the online experiment is much less sensitive to differences in fixation behavior than the in-laboratory experiment because the online experiment does not measure gaze directly. In the laboratory experiment, gaze is measured 350 times per staircase (10 times per trial at 35 trials), whereas the online study only quantified a single number per condition (the threshold). Moreover, gaze is a more direct measurement given our interest in controlling fixation, whereas threshold is only indirectly (and probabilistically) related to fixation behavior.

Discussion

The current study compared the accuracy of stationary fixation, dynamic fixation, and crowded dynamic fixation while participants performed a peripheral task. Gaze accuracy increased from stationary to dynamic to crowded dynamic fixation. The reliable gaze accuracy of crowded dynamic fixation improved our estimates of threshold on the peripheral task, eliminating the abnormally small thresholds observed with the other two fixation tasks. In other words, crowding dynamic fixation greatly reduces the frequency of peeking, and abolishes the abnormally low thresholds that are caused by peeking.

Why does cursor tracking improve fixation?

We instructed participants to manually track the moving crosshair, hoping that they would track it with their eyes. We need the cursor task because we can track the cursor, but not gaze, online. It is natural to look at an object when manually tracking it. Indeed, experiments measuring hand and eye movements show that they are highly correlated during target tracking (Koken & Erkelens, 1992; Xia & Barnes, 1999). Our in-laboratory EyeLink ground truth measurements showed that this works very well, confirming the results from Kurzwski et al. (2023).

The importance of fixation

Accurate knowledge of gaze position, whether by gaze tracking or accurate fixation, is essential for study of many visual tasks. An obvious example is peripheral thresholds, which typically depend strongly on eccentricity. For example, from 0° to 10° eccentricity, grating acuity threshold (degrees per cycle) increases approximately 4-fold (Frisén & Glansholm, 1975), letter acuity threshold (minimal angle of resolution) increases approximately 7-fold (Ludvig, 1941), and crowding distance increases approximately 40-fold (Kurzwski, Burchell et al., 2023). Inconsistent fixation, peeking, produces misleading results: When participants shift their gaze toward a peripheral target, estimated thresholds are profoundly lower (Kurzwski, Burchell et al., 2023).

Location matters for other reasons, too. For example, fixation control is essential in adaptation experiments, where prolonged exposure to a stimulus at a specific visual field location affects perception most strongly at that location (Gibson, 1937). Many studies of object recognition (Potter, 1976) and scene perception (Fei-Fei, Iyer, Koch, & Perona, 2007) ask how much information can be processed from a single glance; in such studies, it is important to know where the participant fixates.

In attention experiments, a peripheral cue improves processing for subsequent stimuli in nearby (but not distant) locations (Carrasco, 2011; Posner, 1980), hence, it is important that participants do not move their eyes during the interval between presentations of cue and target. Similarly, in change-blindness experiments, it is important to know whether participants maintain central fixation or move their eyes to the location of the image where there is a change (Rensink, O'Regan, & Clark, 1997; Simons & Rensink, 2005).

The webcam alternative

Another way to ensure fixation during online vision experiments is webcam-based eye tracking. Saxena, Lange, and Fink (2022) and Papoutsaki et al. (2016) reported that WebGazer, an open source, webcam-based eye tracking tool, measures gaze to within 4°. This is useful for experiments that need dynamic gaze tracking, but is typically not precise enough to ensure fixation for threshold testing. Kaduk et al. (2024) showed that one can improve the gaze precision to 1.5° by adding an external high-resolution webcam. For reference, the gold standard for laboratory-based vision studies is EyeLink 1000, with a precision of 0.5°. However, adding an external webcam limits the study to participants willing to obtain and install additional equipment.

Because our tracking task does not require equipment beyond a typical computer, it can be used at scale for online testing. The experiments presented in this work were conducted in the laboratory to use the EyeLink for ground truth, but with our online web-based software. In addition, previous work by Kurzwski, Pombo et al. (2023) demonstrated that in-laboratory and online testing with EasyEyes yield consistent results.

Does the crowded dynamic fixation task draw more attention than a static fixation mark?

To some extent, typical peripheral testing is always a dual task, because the participant needs to direct their gaze to the fixation mark while anticipating a peripheral target (Pashler, 1994). Does moving and cluttering the fixation mark make the fixation task draw more attention away from the target? We do not have an assessment of the effect of motion on attention, but we did ask four participants to rate the perceived difficulty of tracking the moving crosshair with and without distractors. Admittedly, effort and attention are not quite the same thing. The rated effort was similar for dynamic and crowded dynamic fixation when participants were fixating the crosshair.

We compared crowding thresholds measured with the three fixation tasks. We saw in Figure 6 that

crowding thresholds differ little among the three fixation tasks. To not confound peeking with attention, we re-analyzed those data including only participants who did not peek in any condition (keeping 3 of the 20). Figure 6C, again, shows little difference between thresholds measured with the three fixation tasks. Thus, the crowding thresholds do not differ in a way that could be attributed to differing demands of the three fixation tasks. It would be interesting to assay this again with a task that is more sensitive to attention.

Testing online

The purpose of our study was to assess the effectiveness of crowded dynamic tracking in controlling fixation during peripheral testing. To this end, participants completed the experiment in the laboratory, where precise gaze tracking is available as a ground truth. We made an effort to recruit participants who were not experienced psychophysical observers. Our participants also spanned a broad age range (17–41 years) and were culturally and racially diverse (35% White, 18% Black or African American, and 47% Asian). However, because we recruited our participants locally, it is likely that the sample over-represents WEIRD populations (Henrich et al., 2010; Rad et al., 2018).

Our online experiment tested a broader population, but lacked gaze tracking. We can only infer whether participants peeked during the experiment by examining the crowding thresholds, assuming that peeking is necessary for producing an abnormally low threshold. With stationary fixation, there were many small thresholds. The number of small thresholds was much lower with dynamic fixation, and zero with crowded dynamic fixation. This finding supports our conclusion that the two dynamic tracking methods control gaze, both in the laboratory and online.

Although thresholds can provide some indication of whether participants peeked, they are not a sensitive measure of peeking, and researchers can only use them to exclude participants retrospectively, after the experiment is complete. Alternatively, cursor tracking errors might serve as a proxy for peeking. Although gaze tracking is not routinely available in online settings, our software records both cursor and crosshair position. If errors in cursor tracking reliably predict peeking, we could then exclude or repeat those trials during an experiment. However, by visual inspection of our data, we found no obvious signatures of fixation breaks in our cursor tracking data. A machine learning approach with ground truth gaze data might succeed at identifying fixation breaks from the cursor data alone. However, this would require more labeled data than we currently have.

Conclusions

Kurzwski, Pombo et al. (2023) introduced a manual tracking task for gaze control in psychophysical testing. We improved it by adding independently moving distractors, ensuring gaze remains on the crosshair. Fixation accuracy was verified during peripheral threshold measurement. This task requires no extra equipment and suits online testing. Crowded dynamic fixation provides gaze control for online visual testing, facilitating larger and more diverse samples with minimal effort.

Keywords: fixation, online psychophysics, gaze tracking, EasyEyes, crowding, cursor tracking

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Data and code availability: Raw data and code used to analyze the data reside in <https://github.com/fh986/crowded-dynamic-fixation>.

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Corresponding author: Fengping Hu.

Email: fh986@nyu.edu.

Address: Department of Psychology, New York University, 6 Washington Pl, New York, NY 10003, USA.

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